

VERKHNEIMBATSK, Russian Federation

WMO: 236780

Lat: 63.1555N

Lon: 87.9667E

Elev: 45

StdP: 100.78

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -46.0	(c) -42.6	(d) -48.3	(e) 0.0	(f) -46.1	(g) -45.1	(h) 0.0	(i) -42.7	(j) 6.4	(k) -18.1	(l) 5.8	(m) -17.2	(n) 0.5	(o) 100	(p) 0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	28.0	18.7	26.0	17.9	23.8	17.1	20.0	25.6	18.9	24.1	17.9	22.6	2.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	22.1	17.0	12.2	21.0	15.9	11.4	20.4	57.5	25.7	54.1	24.1	50.8	22.7	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.2	(b) 5.4	(c) 4.7	(e) -47.6	(f) 31.1	(g) 2.7	(h) 1.6	(i) -49.5	(j) 32.2	(k) -51.1	(l) 33.2	(m) -52.6	(n) 34.1	(o) -54.6	(p) 35.3		
(a) 6.2	(b) 5.4	(c) 4.7	(e) -47.0	(f) 22.1	(g) 3.4	(h) 1.8	(i) -49.4	(j) 23.4	(k) -51.4	(l) 24.5	(m) -53.3	(n) 25.5	(o) -55.8	(p) 26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.5	-26.0	-19.5	-10.5	-3.2	4.1	14.3	17.4	13.4	6.7	-2.2	-16.8	-21.3		
	DBStd	16.14	9.96	8.14	7.90	6.58	5.36	5.32	3.92	3.73	4.32	6.09	9.84	10.30		
	HDD10.0	5472	1115	827	635	396	197	21	1	9	118	380	803	970		
	HDD18.3	8049	1373	1061	893	646	441	142	66	159	348	638	1053	1228		
	CDD10.0	528	0	0	0	0	14	149	229	116	19	0	0	0		
	CDD18.3	64	0	0	0	0	1	21	37	7	0	0	0	0		
	CDH23.3	538	0	0	0	0	6	196	293	43	0	0	0	0		
	CDH26.7	114	0	0	0	0	2	43	62	7	0	0	0	0		
(14) Wind	WSAvg	2.3	2.1	2.2	2.5	2.6	2.6	2.2	2.1	2.2	2.3	2.4	2.3	2.2		
(15) Precipitation	PrecAvg	561	36	24	25	32	44	52	59	70	62	65	52	43		
	PrecMax	756	63	51	50	80	96	89	182	127	118	139	84	80		
	PrecMin	373	4	4	2	8	14	14	7	14	16	25	20	19		
	PrecStd	96	14	11	12	15	19	19	35	29	25	25	16	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.6	-1.4	6.6	13.8	23.2	30.2	30.5	27.4	21.6	11.5	2.0	-1.1		
		MCWB	-4.8	-2.1	1.7	6.9	13.8	18.8	20.7	19.5	15.2	9.0	1.6	-1.6		
	2%	DB	-8.5	-4.0	3.2	10.6	18.8	28.0	28.4	24.0	17.9	8.6	0.6	-2.8		
		MCWB	-8.9	-4.7	0.1	4.9	11.0	17.7	19.3	18.0	13.2	6.6	0.0	-3.3		
	5%	DB	-10.7	-6.7	1.0	7.9	15.7	25.7	26.6	21.7	15.3	6.6	-0.5	-5.0		
		MCWB	-11.2	-7.4	-0.9	3.4	9.8	16.8	18.6	16.5	11.7	5.2	-1.1	-5.5		
	10%	DB	-12.9	-8.6	-0.5	5.6	12.7	23.3	24.7	19.6	13.1	4.8	-2.3	-7.3		
		MCWB	-13.4	-9.2	-2.1	2.1	7.7	15.9	17.9	15.5	10.7	3.6	-2.9	-7.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.8	-1.9	2.2	7.2	14.7	20.1	22.5	21.0	15.9	9.6	1.7	-1.6		
		MCDB	-4.5	-1.7	4.9	13.0	21.4	26.8	28.1	26.0	20.0	11.3	2.0	-1.4		
	2%	WB	-8.9	-4.6	0.6	5.3	12.1	19.0	20.6	18.7	13.9	6.9	0.3	-3.3		
		MCDB	-8.6	-4.1	2.6	9.6	17.3	25.3	26.2	22.3	16.9	8.0	0.6	-2.9		
	5%	WB	-11.2	-7.2	-0.7	3.8	10.0	18.0	19.5	17.4	12.3	5.4	-1.1	-5.4		
		MCDB	-10.8	-6.7	1.0	7.1	15.2	24.2	24.9	20.8	14.6	6.5	-0.6	-5.0		
	10%	WB	-13.3	-9.2	-2.1	2.5	8.0	16.8	18.5	16.1	10.7	3.7	-2.8	-7.8		
		MCDB	-12.9	-8.7	-0.6	5.3	12.0	22.2	23.3	19.1	12.9	4.8	-2.2	-7.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.1	9.7	9.3	8.2	9.5	8.9	7.6	6.3	4.8	7.0	6.8		
		MCDBR	7.9	8.2	8.3	10.5	12.6	13.1	12.4	11.3	9.9	5.5	5.7	7.0		
		MCWBR	7.8	8.0	6.5	6.9	7.4	6.2	6.1	6.3	6.1	4.3	5.7	7.0		
		MCDBR	7.8	8.3	7.7	8.8	11.6	10.9	10.8	9.6	7.9	5.2	5.9	7.1		
	5% WB	MCWBR	7.7	8.1	6.6	6.3	7.2	5.9	5.9	5.8	5.5	4.3	6.1	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.289	0.330	0.354	0.369	0.387	0.439	0.412	0.349	0.306	0.231	0.229		
	taud		1.900	1.993	1.988	2.043	2.236	2.374	2.200	2.252	2.403	2.226	1.938	1.869		
	Ebn at Noon		331	607	734	812	841	834	767	753	742	603	343	148		
	Edh at Noon		36	74	112	132	119	107	124	108	76	59	32	18		
(44) All-Sky Solar Radiation	RadAvg		0.24	1.05	2.57	4.33	5.18	5.61	4.94	3.35	1.97	0.79	0.33	0.10		
	RadStd		0.03	0.08	0.20	0.47	0.56	0.48	0.53	0.42	0.27	0.13	0.06	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page